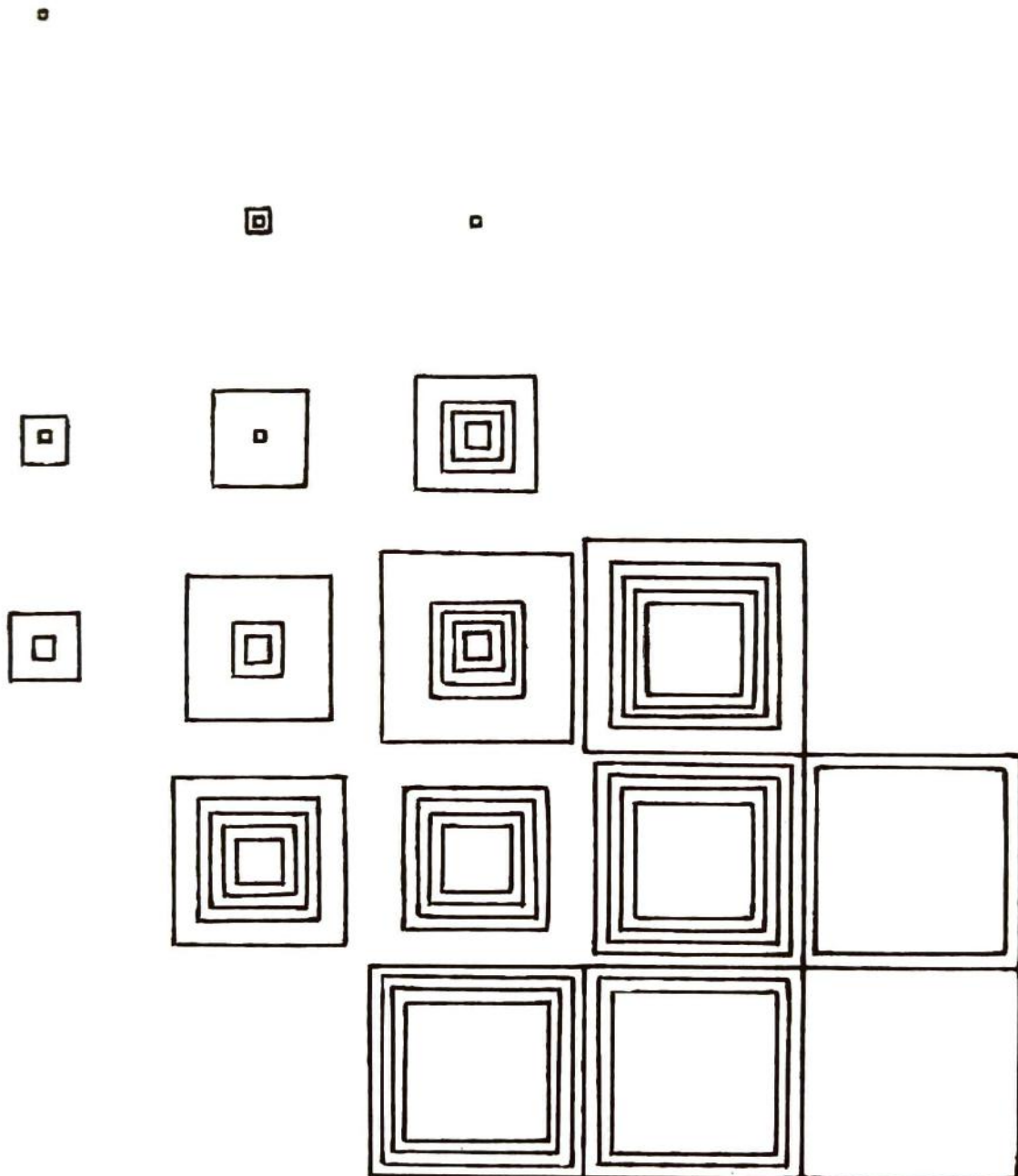


THE MATH SURVEY



....LIFE

Dedication

This, the forty-ninth issue of the Stuyvesant High School Math Survey is proudly dedicated to Mrs. Henrietta Midonick, the retired Chairman of our math department. Mrs. Midonick's dedication and years of service to the school's community is manifest in such concrete examples as the Math Research Lab and in such intangibles as the development and maturity of our entire student body.

THE MATH SURVEY

PUBLISHED BY THE MATHEMATICS DEPARTMENT OF STUYVESANT HIGH SCHOOL

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Editor's Note

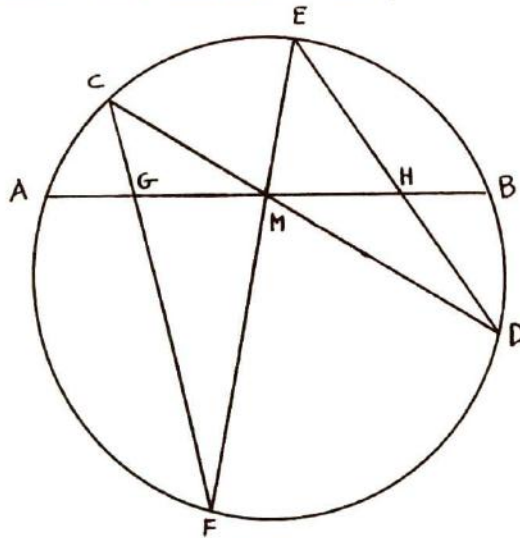
The Math Survey that you are now reading traces its lineage back forty-nine years, to 1926, when the Mathematics Department inaugurated what was to become Stuyvesant's finest and only self-sufficient magazine. This issue represents indeed hundreds of hours of dedicated work by our small but diligent staff. In addition to them, we would like to express our thanks to Mr. Barry M. Glotzer, our faculty advisor, who, despite becoming the proud father of a baby girl Lisa Yvette, found the time to help, guide and trade criticisms with us.

Ever cognizant of our charge to constantly improve Survey, we are proud to note that this is the largest issue ever. The articles are on topics never before discussed in Survey and rarely treated in most math magazines. In addition to articles, there are mathematical notes presenting novel or new proofs or useful techniques. The problem section is the largest ever and includes a collection of research problems.

As the fiftieth anniversary of Survey approaches, we wish to rededicate this magazine to "real mathematics" the beautiful math that lies just beyond the high school classroom that we hope every Stuyvesantian will someday find and appreciate.

Butterfly Problem

This famous problem appeared on the inside cover of the 1972 Math Survey with its usual complicated proof making use of the construction of several auxiliary lines and circles. Most recently a simple solution appeared. It is given on page 32. Try it first, it is a good challenge. (Hint: Try using areas.)



Given: $\overline{AB}$ an arbitrary chord of a circle. M is the midpoint of $\overline{AB}$ and $\overline{CD}$ and $\overline{EF}$ are chords through M. $\overline{CF}$ and $\overline{ED}$ intersect $\overline{AB}$ at G and H, respectively.
Prove: $GM = MH$

Formula for π by Paul Zeitz

Since Archimedes, mathematicians have been devising methods of calculating π . One of the most well known methods is Leibnitz's series:
 $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + 1/9 - \dots$. Most mathematical treatments of this series prove it with the aid of the calculus as follows:

$$\pi/4 = \arctan 1 - \arctan 0 = \int_0^1 dx/(x^2+1) \quad (\text{since } dx/(x^2+1) = \arctan x) = \sum_{i=0}^{\infty} (-1)^i x^{2i+1}/(2i+1) = 1 - 1/3 + 1/5 - 1/7 + 1/9 - \dots$$

This proof is fairly straightforward, if you know the calculus! However, there is a little known proof of Leibnitz's series that involves very little in the way of technical prerequisites. There are a few concepts that may not be known to the reader which I will list here.

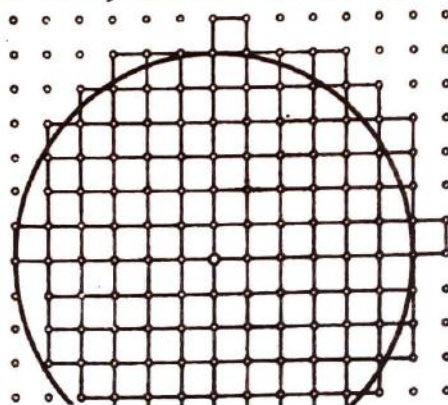
1. $[n]$ denotes the greatest integer less than or equal to n . For example, $[3]=3$, $[\pi]=3$, $[2.6]=2$, $[-.25]=0$. It is easy to see that, given two integers m , and n , the number of integers from 1 to n inclusive having m as a factor is $[n/m]$.

2. The equation $x^2+y^2=n$, where x, y , and n are all integers, may or may not have a solution depending on n . The number of solutions (possibly zero) is equal to 4 times the difference of the number of factors of n of the form $4k+1$ and the number of factors of n of the form $4k+3$. This includes both positive and negative solutions. For example, let $n=45$. The factors of the form $4k+1$ of 45 are 1, 5, 9, 45. The factors of the form $4k+3$ are 3, 15. The difference is $4-2=2$. Thus the number of solutions is 8. These solutions are $(6,3)$, $(-6,3)$, etc. and $(3,6)$, $(-3,6)$, etc. Those persons interested in a proof of this could consult chapter 5 of Niven and Zuckerman's An Introduction to the Theory of Numbers. The proof is fairly complicated and there is no room to include it here.

The geometrical device that we shall use in our proof is the notion of a lattice point in coordinate geometry. A lattice point is merely a point in the plane with integer coordinates.

On our plane of lattice points, we draw a circle with radius $\sqrt{n}$. The equation of this circle is of course $x^2+y^2=n$. We form the function $r(n)$, where $r(n)$ is equal to the number of lattice points on the circle with radius $\sqrt{n}$. (We are assuming that n is an integer.) We know that $r(n)$ is equivalent to the number of solutions in integers to $x^2+y^2=n$, a formula we already know (see 2.) If we then form the function $R(n) = \sum_{k=0}^n r(k)$ ($r(0)$ is defined to be 1), we will have a new function, equivalent to the number of lattice points on or in the interior of our circle. We can see intuitively that $R(n)$ is approximately equal to the area of the circle, or πn , since to each lattice point we can assign a unit square with unit area. Thus, $R(n)/n$ is an approximation of π . The greater n becomes, the closer $R(n)/n$ comes to π , and in fact we have $\lim_{n \rightarrow \infty} R(n)/n = \pi$.

Proof: Let us draw our circle, (see figure), and for each lattice point on or inside the circle we assign a different unit square such that our lattice point is the lower left hand corner of the square. Now the difference between $R(n)$ and the area of the circle, πn , is at most equal to the area of the squares (in the diagram)



that are cut by the boundary of the circle. We shall call this area $A(n)$. Hence we have $\left| \frac{R(n)}{n} - \pi \right| < \frac{A(n)}{n}$. Now, the total

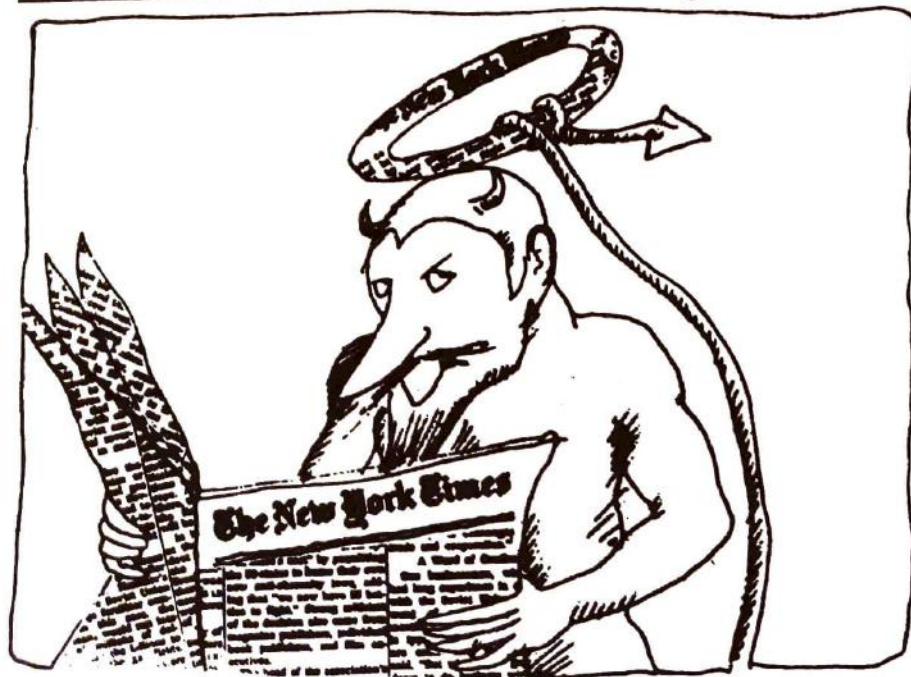
possible area for all squares cut by the boundary of the circle would be less than the area of the circular shell formed by the radii $\sqrt{n} + \sqrt{2}$, and $\sqrt{n} - \sqrt{2}$. (Since the length of the diagonal of a unit square is $\sqrt{2}$). We shall call this area $B(n)$, or $B(n) = \pi[(\sqrt{n} + \sqrt{2})^2 - (\sqrt{n} - \sqrt{2})^2] = 4(\sqrt{2})\pi$. Of course $A(n) < B(n)$, therefore $\left| \frac{R(n)}{n} - \pi \right| < \frac{4(\sqrt{2})\pi}{n}$. Letting n increase

without limit we will have $R(n)/n \rightarrow \pi$ as $n \rightarrow \infty$. Q.E.D.
With this result we can now prove $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$

Proof: First we will show that $\frac{1}{4}(R(n)-1) = \sum_{i=1}^n (-1)^i [n/2i+1]$. Then, by letting n increase without limit, we can arrive at Leibnitz' series. We know that $R(n) = \sum_{i=1}^n 4N(i)$, where $N(i)$ represents the difference between the number of factors of i of the form $4K+1$ and the number of factors of i of the form $4K+3$. Since $4N(0)=1$, we get $\frac{1}{4}(R(n)-1) = \sum_{i=1}^n N(i)$. Now we must find the sum. We list the integers of the form $4k+1$: 1, 5, 9, 13, 17, ... Now, there are n integers from 1 to n with the factor 1, hence they are counted n times. However, there are only $n/5$ integers between 1 and n with the factor of 5, hence they are counted $n/5$ times. Applying this argument to all integers of the form $4k+1$ and also of the form $4k+3$, we get $\frac{1}{4}(R(n)-1) = n + n/5 + n/9 + n/13 + \dots - n/3 - n/7 - n/11 - \dots$, and by changing the order of the terms we get $(R(n)-1) = n - n/3 + n/5 - n/7 + n/9 - \dots$. This series has $[\frac{1}{2}(n+1)]$ terms. Let us assume that n is an odd perfect square. If we end the series after the term $[n/\sqrt{n}] = \sqrt{n}$, since the absolute value of each term is decreasing, the total error will be $k\sqrt{n}$, where $k < 1$. Then, if we drop the brackets completely, since the absolute value of each term is < 1 the total error would be $j\sqrt{n}$, where $j < 1$. Therefore, we now have $(R(n)-1) = n - n/3 + n/5 - n/7 + n/9 - \dots + k\sqrt{n} + j\sqrt{n}$.

Dividing by n and letting n increase without bound, we arrive at our result. Q.E.D.

For those who found the foregoing interesting, I strongly recommend reading Geometry and the Imagination, by D. Hilbert and S. Cohn-Vossen. The book contains an interesting section on the applications of lattice points, among other things.



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Vector Geometry

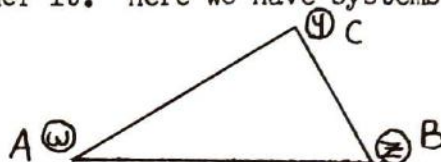
by Jesse Deutsch

We shall prove some propositions in Euclidean geometry by means of vectors and mass points. The theorems are Ceva's, Menelaus', Varignon's parallelogram, Desargues' and Euler's.

Some assumptions that we make are that the center of mass of a system is unique, and that vectors obey the commutative and associative laws under addition.

Mass Points

The center of mass of a system is that point where the system would balance if a pin were to be placed under it. Here we have systems of two and three masses, respectively:



The law of the lever tells us that $wx = Wy$ or $w/W = y/x$. To find the center of mass of the triangle we must first find the center of mass of any two of its vertices, say A and C, and then locate the center of mass of the line joining this point to the third vertex B. The uniqueness of the center of mass tells us that all three lines, from a vertex to the center of mass along the opposite line, intersect in one point.

Using the law of the lever on the diagram at the right we have,

$$\frac{AD}{DB} = \frac{z}{w}, \frac{EC}{AE} = \frac{w}{y}, \frac{BF}{FC} = \frac{y}{z}$$

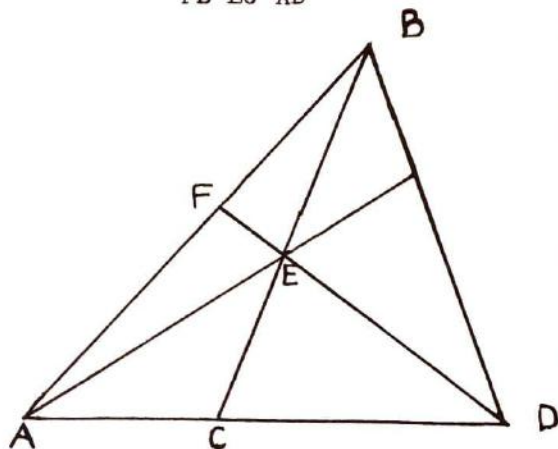
so that:

$$\frac{AD}{DB} \cdot \frac{EC}{AE} \cdot \frac{BF}{FC} = 1$$

which is Ceva's Theorem.

We can obtain Menelaus' Theorem in a similar manner. It states that if line FED intersects $\triangle ABC$, F being on AB, D on AC, E on BC then:

$$\frac{AF}{FB} \cdot \frac{BE}{EC} \cdot \frac{CD}{AD} = 1$$



$$\text{Hence, } \frac{AF}{FB} \cdot \frac{BE}{EC} \cdot \frac{CD}{AD} = \frac{b}{a} \cdot \frac{a+d}{b} \cdot \frac{a}{a+d} = 1$$

Since $\overline{AG}$, $\overline{BC}$, $\overline{FD}$ intersect at E we may consider E as the center of mass of $\triangle ABD$ appropriately weighted, as on the left. (Note that the weight at C is given by the sum of the weights at A and D.) Then:

$$\frac{AF}{FB} = \frac{b}{a} \quad \frac{BE}{EC} = \frac{a+d}{a}$$

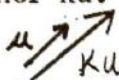
$$\text{Now, } \frac{AD}{CD} = \frac{CD+AC}{CD} = 1 + \frac{AC}{CB} = 1 + \frac{d}{a}$$

$$\text{Then, } \frac{CD}{AD} = \frac{a}{a+d}$$

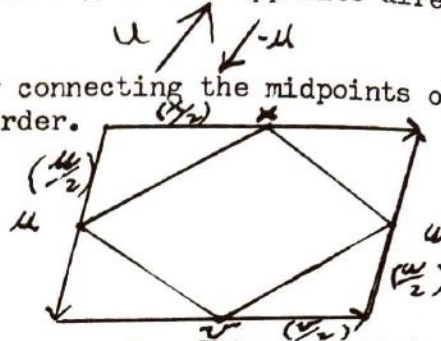
Similar means can be used in the case where $\overline{FED}$ intersects one of the sides of the triangle extended.

Vectors

Vectors have a magnitude and direction. Two vectors are equal only if they have the same direction and length (magnitude). For two vectors that are parallel having lengths in the ratio of $1:k$, we could call one $\vec{u}$, and the other $k\vec{u}$. Vectors with opposite signs have opposite directions



Consider the quadrilateral formed by connecting the midpoints of the sides of an arbitrary quadrilateral, in order.



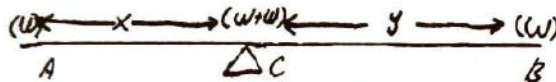
Then $\vec{u} + \vec{v} + \vec{w} + \vec{x} = \vec{0}$

Also $x/2 + a = u/2$ or $a = (u-x)/2$

And $v/2 + w/2 = b$

Therefore $a + b = (u-x + v+w)/2 = \vec{0}$, or $a = -b$. This says that a is parallel to b and as long but oppositely directed. Hence, the inscribed quadrilateral must be a parallelogram (since we can also apply the same argument for the other sides). This result is known as Varignon's Parallelogram.

Going back to the lever we see the following



$$wx = Wy$$

The weight at A is w , at B is W , and at C is $w+W$. Since C is that unique point where the system balances, we can say that

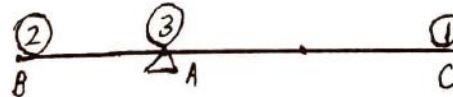
$$wA + WB = (w+W)C$$

Conversely, if we know that there are w, W such that $wA + WB = (w+W)C$ then C must be on line segment AB.

What if one of the weights is negative? Say for example that we have $3A - 2B = C$ or $3A = 2B + C$ and the figure looks like so that C is not on segment AB but somewhere on line AB.

Notice that if we use vectors

$$\frac{AB}{AC} = \frac{-BA}{AC} = \frac{-1}{2}, \quad \frac{AC}{CB} = \frac{-2}{3}$$



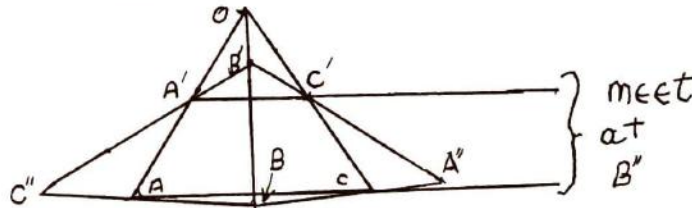
Thus given $aA + bB = (a+b)C$ we can define $\frac{AC}{CB} = \frac{b}{a}$ and this will make sense

whether C is between A and B or not.

We will need negative weights for we will use them in proving

Desargue's Theorem:

OAA', OBB', OCC' , are collinear. Show that $A'', B'',$ and C'' are collinear.



We know that

$$O = mA + nA' = pB + qB' = rC + sC' \quad \text{with } m+n=1, p+q=1, \text{ and } r+s=1$$

or $mA - pB = qB' - nA' = (m-p)C''$ where C'' must necessarily be the intersection of line AB and line $A'B'$. We also find

$$pB - rC = (p-r)A''$$

$$rC - mA = (r-m)B''$$

adding

$$O = (m-p)C'' + (p-r)A'' + (r-m)B''$$

or

$$(m-r)B'' = (m-p)C'' + (p-r)A'' \quad m-p + p-r = m-r$$

so B'' is on line $A''C''$,

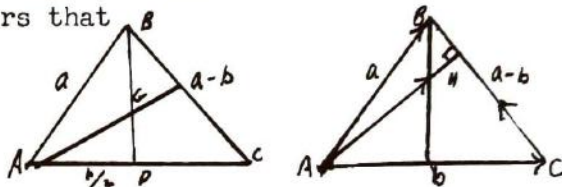
We know from elementary geometry that the medians of a triangle intersect at a unique point called the centroid. Similarly the altitudes intersect at a point called the orthocenter. The Euler line theorem states that the centroid, the orthocenter, and the center of the circle circumscribed about the triangle all lie on the same line.

To prove the theorem we must first introduce the concept of dot product. If a and b are two vectors, then $a \cdot b = \|a\| \|b\| \cos \theta$ where $\|a\|$ is the metric length of vector a , and θ is the angle between vectors a and b . It can be easily shown that the dot product obeys the following laws:

$$a \cdot (b+c) = a \cdot b + a \cdot c \quad a \cdot b = b \cdot a \quad a \cdot a = \|a\|^2 \quad a, b, c \text{ are arbitrary vectors.}$$

Let G be the intersection point of the medians of triangle ABC and let vector AB be called ' a ', vector AC be called ' b '. From Ceva's Theorem and other work with vectors that

$$\begin{aligned} BG/GD &= 1/2 \\ BG &= -(a-b)/2 \\ DG &= (a-b)/2 \\ AG &= b/2 + a/3 - b/6 \\ &= (a+b)/3 \end{aligned}$$



If we call the intersection of the altitudes H , then we know the following:

$$\begin{aligned} (a-v) \cdot b &= 0 \\ (a-b) \cdot v &= 0 \text{ or} \\ a \cdot b &= a \cdot v = b \cdot v \end{aligned}$$

It is easy to show that $a \cdot b = 0$ implies a is perpendicular to b , and vice versa.

Let O be such that $HG:GO = 2:1$, then

$$\frac{(v-x)}{(a+b)/3-x} = \frac{2}{1}$$

or, $a+b-3x=v-x$, $(a+b-v)/2 = x$

Now we must show that x is equidistant from the vertices.

$$\begin{aligned} x^2 &= x \cdot x = (a \cdot a + b \cdot b + v \cdot v + 2a \cdot b - 2a \cdot v - 2b \cdot v)/4 \\ (a-x)^2 &= (a+x)(a-x) \quad \text{now, } a-x = a - (a+b-v)/2 = (a+v-b)/2 \end{aligned}$$

$$\text{Hence, } (a-x)^2 = (a \cdot a + b \cdot b + v \cdot v + 2a \cdot v - 2a \cdot b - 2v \cdot b)/4$$

$$\text{Also, } (b-x)^2 = (a \cdot a + b \cdot b + v \cdot v + 2a \cdot v - 2a \cdot b - 2v \cdot b)/4$$

$$\text{Then } x^2 = (a-x)^2 = (b-x)^2 \text{ and } x = a-x = b-x.$$

So the point O is equidistant from the vertices and is hence the circumcenter.

For more information about vectors and mass points read either A Vector Space Approach To Geometry by Melvin Hausner or Geometry Revisited by H. Coxeter and S. Greitzer



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Reflections and Projections

by Edward Marcantonio, Chairman Math Dept.

My many years of teaching experience convince me that Stuyvesant is a most unusual high school. It is a place where all students are highly motivated by a desire to learn. They actively seek opportunities to devour the knowledge that civilization has made available and then strive to contribute to it. Stuyvesant is a place where the absence of a regular teacher allows a gifted student to deliver a fascinating lecture on the theory of numbers involved in an original research project on "quasi-perfect numbers". It is a place where a winning project in the Westinghouse competition includes the fact that a Venus Flytrap exemplifies the concept of a direct variation in mathematics. The atmosphere here is one of intellectual excitement. Students provide catalytic action for one another and they are made to feel comfortable in that role. Too often in a less academic atmosphere bright students resort to silence and retreat from opportunities to add new learnings lest they incur the disapproval of their classmates. Our comfortable academic surroundings actually inspire pupils to develop qualities of leadership. In the Mathematics Department they play a major role in training and conducting the affairs of a championship mathematics team. The student leadership of the Mathematics Society arranges an after school lecture series that exposes our students to some of the finest mathematical minds of our country.

With all this student talent we feel a strong obligation to continue to provide pupils with an exciting climate for intellectual growth. Student maturity and leadership talents must be recognized and utilized in promoting our school's program. We have been receptive to the requests of pupils for new electives in advanced computer science, number theory and the history of mathematics. It is our hope that the new Research Resource Center will be in operation throughout the school day. Resource teachers will then be available to assist students in the preparation of term papers and research papers for the Westinghouse and other competitions. Some pupils may wish to use this facility to pursue independent study and possibly obtain advanced placement credit in a mathematics course. Or perhaps the center will provide a chance for pupils to get answers to questions of mathematical importance.

Most of our students choose to attend Stuyvesant because they have particular aptitude in mathematics and science. If we are to continue with our specialized status we must be able to justify ourselves as a place where young people can develop excellence in these areas.

To achieve this goal I would like to see every pupil take at least 4 years of high school mathematics. In fact,

it is my hope that this will become a Stuyvesant graduation requirement. I am distressed when I learn that some pupils do not take mathematics electives as seniors because they desire a "short program in order to take it easy in their last semester". Yet, in the same breath these students profess a great desire to attend a prestigious college, a college made famous by the great demands it places upon the time and minds of its students. This semester we hope to institute a series of guidance sessions conducted by the Mathematics Department whereby the offerings in mathematics will be explained and the advantages of electing our courses will be cited. I have found that young people desire guidance and are receptive to suggestions which will better prepare them for the future.

Perhaps outstanding evidence of the opportunities we offer for the development of student initiative and leadership is the publication of Math Survey. I have noted with pride the efforts of our pupils and faculty advisor, Mr. Glotzer as they carried this edition from the planning stage to publication. Our congratulations to the staff for this fine edition. It has captured the spirit of our department and speaks well of us.

Research Laboratory

This year for the first time Stuyvesant High School opened its resource center for the study of mathematics. Also called the mathematics research lab, it exists primarily because of the devotion and dedicated work of Mrs. Henrietta Midonick, our former Math Dept. chairman, and Mr. Marcantonio, our present chairman, with the help of Mrs. Finkel, Mr. Glotzer, and Mrs. Tunis. This term it is being staffed by Mrs. Tunis during sixth period. The center has a library of mathematics books, to be used by students for research, term papers, papers for competitions such as the Metropolitan Math Fair, and also just plain reading. Also available are various office equipment for the use of students to assist in the writing of such papers.

With such facilities the resource center is an excellent place for students to get help with a difficult math problem, to research some interesting term papers, and to meet others with an interest in math. The resource center was financed through a state grant.



156 SECOND AVENUE
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NEW YORK, N. Y. 10003

Computer Society

by Cole Whiteman

The Stuyvesant High School Computer Society consists of the operators and trainees who make use of our I.B.M. 1130 and H.P. 9100B computers. In addition to running our system for the computer science classes, the club members train others in operation, give minicourses in computer science and attempt to improve the operating system under faculty supervision.

For example, this year we are blessed with a new version of the FORTRAN programming language for the I.B.M. 1130 that was developed by former Stuyvesant operators now attending college. It supports logical and complex variable types. The 'operator communications package' (affectionately known as GRIN) which enables direct interaction with the computer from the console keyboard was composed a few years ago by a team of operators.

This spring, three minicourses in computer science were given by Society members during the April 3-5 minicourse session. These courses were "Understanding DM3- a guide to the I.B.M. 1130 " by Society President and Chief Operator Nordechai Beizer, " a Primer in PL/I" by Cole Whiteman which was an introduction to a programming language more flexible and powerful than FORTRAN and "APL/1130" by Sidney Gudes introducing a mathematical interactive language that is extremely concise and easy to learn.

Our assistant chief operators are Sidney Gudes and Louis Obando. Other operators include Brad Albom, George R. Hess, George Hruza, William Scott, Fred Topol and Cole Whiteman. The faculty advisor is Mr. J. Rosenrauch.

Summation

Among the many abstruse and esoteric notations used in mathematics is Σ , "the sigma sign" meaning summation. Usually written as $\sum_{i=1}^n f(i)$ it simply means to add up all the successive values that $f(i)$ takes on as i is incremented from one to n by ones (as $i=1,2,3,\dots,n$). It is read as "the summation of $f(i)$ as i goes from 1 to n ". The sigma notation is used in many places because it is elegant and easy to write. An example will show the usefulness of the symbol.

$$\sum_{i=1}^n i = 1 + 2 + 3 + \dots + i \dots$$

A student familiar with progressions will see that we can represent them in this notation.

$\sum_{i=0}^{n-1} (ar^i - 1) = a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$ is the standard geometric progression and

$\sum_{i=0}^n (a+id) = (a) + (a+d) + (a+2d) + \dots + (a+nd)$ is the arithmetic progression.

If we let n get larger and larger it approaches ∞ then we have the infinite series. One of the more interesting infinite summations is $\sum_{k=1}^{\infty} (1/k^2)$ which turns out to be $\pi^2/6$.

A nice series is $\sum_{i=1}^n i = n(n+1)/2$. Try and prove it!

Students will generally find the summation used throughout mathematics and it is very useful to become acquainted with it.

Mathematics Society by Kelly Pan

This year has been one of the busiest for the Stuyvesant High School Math Society. We have been able to attract many highly qualified professors and lecturers from such institutions as Princeton University, Columbia University, New York University, Hampshire College in Amherst, and even New College in Sarasota, Florida. We had fantastic success with our first lecturer from the Stuyvesant Math Department, Mr. Glotzer on "Polyhedra". Among the lectures this year were "Theory of Cardinal Numbers", "The Projective Plane", "Number Theory", "Geometry", "probability" and "It's Snowing on the Number line" introducing measure theory. Our plans have not been restricted to pure math alone. We held a joint meeting with the Astronomy Society for a lecture on the mathematical aspects of "Current Topics in Astrophysics". In the future we expect to expand our society by lectures on applied math and have some student lecturers.

Society President Kelly Pan and Vice-President Cole Whiteman with the help of faculty advisor Mr. K. Ritterman and active society members Jesse Deutsch, Richard Pan, Yuval Gilbert, Alex Theodorou and Paul Zeitz have prepared our busy schedule. Eric Lander and Steven Kerstein also were especially responsible for recruiting lecturers.

The Math Society tries to present students with mathematics not normally encountered in the high school classrooms. It is hoped that we have and will continue to achieve this goal.

BEST WISHES TO
MATH SURVEY FROM THE

STUYVESANT HIGH SCHOOL
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Mod x

When discussing time, we are interested in the number of hours that have passed since the last occurrence of twelve o'clock. It would be too cumbersome to say, "It's 148 since noon last Sunday." Hence, we discard multiples of twelve from our reckoning. (The army works on a similar system, but uses 24 hours, instead.) It is logical that a theory would develop around the remainders that numbers leave when divided by other numbers.

Def: We say " $a \equiv b \pmod{m}$ " for integers a and b , read " a is congruent to $b \pmod{m}$ " if the difference $a-b$ is a multiple of m . (This is equivalent to saying that a and b leave the same remainder on division by m).

Verify: $1 \equiv 3 \pmod{2}$
 $17 \equiv 5 \pmod{4}$
 $17 \equiv 5 \pmod{3}$
 $-2 \equiv 15 \pmod{17}$
 $3 \equiv -2 \pmod{5}$

The " $\equiv$ " symbol gives us a new relation, akin to $=$ (actually, it gives us an infinite number of relations, congruence mod m , for each integer m .) The " $\equiv$ " acts very much like " $=$ ". Observe:

THM: If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

- (1) $a+b \equiv c+d \pmod{m}$, $a-b \equiv c-d \pmod{m}$
- (2) $ab \equiv cd \pmod{m}$

proof: $a \equiv b \pmod{m}$ means that $a-b=km$ or $a=b+km$. Similarly, $c=d+jm$. Hence, $a+c=b+d+(j+k)m$ or $(a+c)-(b+d)=(j+k)m$. But, this is exactly the condition that $(a+c) \equiv (b+d) \pmod{m}$. The proofs of the remaining statements are similar.

Division cannot be easily handled, since we are restricted (see definitions) to integers. Many problems arise. For instance, $2x \equiv 0 \pmod{6}$ does not necessarily imply that $x \equiv 0 \pmod{6}$, since $x \equiv 3 \pmod{6}$ works as well.

We can, however, say the following.

THM: The equation $ax \equiv b \pmod{m}$ has a unique solution iff (if and only if) a and m are relatively prime (share no factor).

proof: is omitted, but the idea is important. (HINT: $1=ax+my$ therefore $b=bax+bmy$.)

One possible use of mods is demonstrated in the solution of the following problem:

In a fight between the septacaptians (seven-headed beasts) and the decasepta-tetes (seventeen-headed creatures) an observer's head-count at the height of the battle totaled 86 heads. How many creatures were involved?

solution: Let x and y denote, respectively, the number of 7 and 17 headed creatures in the fight. Then $7x+17y=86$.

If two things are equal then surely they are congruent in any mod, specifically; $7x+17y \equiv 86 \pmod{7}$.

Now, $7 \equiv 0 \pmod{7}$, $17 \equiv 3 \pmod{7}$, $86 \equiv 2 \pmod{7}$. With these replacements we have: $3y \equiv 2 \pmod{7}$

Multiply by 5 (which is "3-inverse"), noting that $3 \cdot 5 \equiv 1 \pmod{7}$
Hence: $5 \cdot 3y \equiv y \equiv 5 \cdot 2 \equiv 3 \pmod{7}$

We then should take $y =$ some number that is congruent to 3 mod 7. Try $y=3$. Then $x=7$. (It turns out, that for any larger suitable y the corresponding x value would be negative. BAD!!)

The applications of mods are many and well distributed throughout many areas of mathematics. This short introduction should pave the way for deeper and more interesting work that is available on mods.

Math Team

Eric S. Lander and Jesse Deutsch

The Math Team is a well-ordered subset of the Math Department which meets daily at 8AM in room 517 to prepare for city, state, and national competitions. With the coaching of Mrs. Finkel, captain Eric Lander and co-captain Jesse Deutsch have led one of the best teams in the school's history to victory.

In the New York City Interscholastic Mathematics League Fall Competition, Stuyvesant's score of 111 left the rest of the city far behind (closest was an 82). Scores of 27 for Eric and Jesse and 26 for junior Paul Zeitz allowed Stuyvesant to capture all the individual scoring medals for the city. Also on the team are seniors Kelly Pan and Francis Barany.

For the first time, Stuyvesant fielded a junior team in the IML which placed a strong second in the city. With the experience they acquired, hopes run high for the spring term.

In the National Mathematics Association of America exam, Stuyvesant's score of 385 placed us at the top of the nation. Individual high scores included 145 for Eric Lander, 128 for Paul Zeitz, and 112 for Jesse Deutsch. These three students participated in the 3rd USA Mathematical Olympiad.

On May 18, two fifteen member teams (including the ENTIRE Stuyvesant Senior Team) representing New York City, took first and third place in the New York State Mathematics League Competition held at New Paltz, retaining for NYC the title of Math Capital of the World won last year.

Below we have reproduced some problems from a recent Math Team competition:

IML Competition. (12 minutes for each pair)

1. Find three consecutive odd integers the sum of whose squares can be represented by a four-digit decimal numeral consisting of four identical digits.

2. Three non-zero numbers are in arithmetic progression. The progression can be converted into a geometric progression either by adding 1 to the first term or by adding 3 to the third term. Find the 2nd term.

3. Find all integral values of x and y that satisfy
$$x^2 - 16x + 63 = 2^y.$$

Express your answer in the form (x,y) .

4. In the square ABCD, point E is on DA and the perpendicular to EC at C intersects AB at F. Find the ratio of DE:DA if the ratio of the area of region ECF to that of region ABCD is 5:2.

5. As x varies over all real numbers, what is the minimum value of the expression $|x-5| + |x-3|$?

6. Given $m\angle AOB = 30$ and the inch-length of OA is 12. The foot of the perpendicular from A to OB is A_1 . The foot of the perpendicular from A_1 to OA is A_2 . The foot of the perpendicular from A_2 to OB is A_3 . Continuing in this way, we obtain A_4, A_5 , etc., without end. The infinite sum $AA_1 + A_1A_2 + A_2A_3 + \dots = 24 + \sqrt{k}$. Find k .

Two Views of Fermat's Little Theorem

by Kelly Pan

The aim of this article is to prove Fermat's Little Theorem in two ways -- number theoretic and group theoretic. At the same time we will introduce some basic ideas of both fields. This theorem says that $(a^{p-1}-1)$ is divisible by a prime p if a is not divisible by p . It was Euler, however, who supplied the proof for this theorem since Fermat following his usual course did not put down his proof.

First we shall need some concepts from number theory several of which can be found in the Mathematical Note on Mods in this issue. Congruences have many properties similar to equality. We can prove that it is an equivalence relation, that is the relation is reflexive, symmetric, and transitive.

- 1) reflexive--for any integer, $a \equiv a \pmod{m}$ because $m/(a-a)$.
- 2) symmetric--if $a \equiv b \pmod{m}$ then $b \equiv a \pmod{m}$. If $m/(a-b)$, then $m/-(a-b)$ which means $m/(b-a)$ or $b \equiv a \pmod{m}$.
- 3) transitive--if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $a \equiv c \pmod{m}$. If $m/(a-b)$ and $m/(b-c)$ then $m/((a-b)+(b-c))$ which says $m/(a-c)$ or $a \equiv c \pmod{m}$.

We shall also need some definitions on residues:

Def: When two numbers are congruent, each is a residue of the other modulo (mod) m .

Def: All integers congruent to a given integer mod m compose a residue class.

The numbers in a residue class are distinct from all the numbers in other residue classes. It can be seen that there are m distinct residue classes which can be represented by the least positive numbers as $0, 1, 2, \dots, (m-1)$.

Def: $\phi(m)$ is the notation for the number of numbers less than n and relatively prime to m .

Def: A reduced system of residues mod m is made up of residue classes of integers relatively prime to m . Then there are $\phi(m)$ numbers in each reduced system of residues.

With these definition and properties in our knowledge we can proceed to prove Fermat's Theorem.

Proof: Consider the reduced system of residues mod m

$$p_1, p_2, p_3, \dots, p_{\phi(m)}.$$

Let a be a number relatively prime to m . Then another reduced system is

$$ap_1, ap_2, ap_3, \dots, ap_{\phi(m)}.$$

Therefore the two systems are congruent to each other in some order, that is for each p_i there is congruent to it some ap_j where $i, j < \phi(m)$. Multiplying them together and setting them equal

$$a^{\phi(m)} p_1 p_2 \dots p_{\phi(m)} \equiv p_1 p_2 \dots p_{\phi(m)} \pmod{m}.$$

Cancelling out by the product $p_1 p_2 \dots p_{\phi(m)}$ since it is relatively prime to m , we get a more general version of Fermat's theorem,

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

In the special case where m is a prime p , then $\phi(m) = p-1$ we get the little theorem $a^{p-1}-1$ is divisible by p .

Now we can proceed to prove Fermat's Little Theorem by group theoretic means.

A group G is a nonempty set of elements with a defined binary operation $\circ$ called a product such that the following properties hold.

- 1) Closure- $a, b \in G$ implies that $a \circ b \in G$.
- 2) Associative- if $a, b, c \in G$ then $a \circ (b \circ c) = (a \circ b) \circ c$.
- 3) Identity- $\exists$ a single element $e \in G$ such that $a \circ e = e \circ a = a$ for all $a \in G$.
- 4) Inverse- for all $a \in G$ $\exists$ a unique element $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.

There is one major theorem which we shall need and which shall be proved with the help of some lemmas and definitions.

Lemma 1 $(a \circ b)^{-1} = b^{-1} \circ a^{-1}$

Proof: $e = (a \circ b) \circ (b^{-1} \circ a^{-1}) = a \circ (b \circ b^{-1}) \circ a^{-1}$ (by the associative property)
 $a \circ e \circ a^{-1} = e$. Therefore $(a \circ b)^{-1} = b^{-1} \circ a^{-1}$.

Lemma 2 $(a^{-1})^{-1} = a$

Proof: $a^{-1} \circ (a^{-1})^{-1} = e = a^{-1} \circ a$

Lemma 3 Given a, b in group G then the two cancellation laws hold

- 1) $a \circ u = a \circ w$ implies $u = w$
- 2) $u \circ a = w \circ a$ implies $u = w$

Proof: $a \circ u = a \circ w$

$\exists b \in G$ such that $b \circ a = e$

$b \circ (a \circ u) = b \circ (a \circ w)$

$u = e \circ u = (b \circ a) \circ u = b \circ (a \circ u) = b \circ (a \circ w) = (b \circ a) \circ w$

$u = (b \circ a) \circ w = e \circ w = w$

The same proof is used for the cancellation on the right.

Def: A subset H of group G is a subgroup if H itself is a group under the binary operation in G .

Lemma 4 H is a nonempty subset of group G and H is closed under multiplication. Then H is a subgroup of G . This lemma is relatively easy to prove and will be left to the reader because of lack of room.

Def: If H is a subgroup of group G and $a \in G$, then $Ha = \{ha \mid h \in H\}$ is called the right coset of H in G .

Def: the order of G , $o(G)$, is the number of elements in G .

Def: $a \equiv b \pmod{H}$ if $ab^{-1} \in H$ [this is similar to congruences in number theory]

Lemma 5 for all $a \in G$, $Ha = \{x \in G, a \equiv x \pmod{H}\}$

Proof: let $[a] = \{x \in G, a \equiv x \pmod{H}\}$
 We will first show that $[a] \supset Ha$.

$h \in H$, $a(ha)^{-1} = a \circ a^{-1} \circ h^{-1} = h^{-1} \in H$ implies

$ha \in [a]$ for all $h \in H$ and so $[a] \supset Ha$.

Now suppose that $Ha \supset [a]$ or $ax^{-1} \in H$

$(ax^{-1})^{-1} = xa^{-1} \in H$ implies $xa^{-1} = h$, $h \in H$

$x = ha$ implies $x \in Ha$ which means $Ha \supset [a]$.

Having proved that $Ha \supset [a]$ and $[a] \supset Ha$ this means that $Ha = [a]$.

We have just proved that Ha is a residue class but that means these classes divide G into distinct, disjoint sets. Any two right cosets of H in G are either the same right coset or have no common elements.

Lemma 6 $\exists$ a 1-1 correspondence between any two right cosets Ha and Hb of subgroup H in group G . This simply means that any two right cosets have the same order and the order of any right coset of H is $o(H)$.

Proof: If $h_1b=h_2b$ then $h_1=h_2$. The order of the right coset He , $o(He)=o(H)$. Therefore all the right cosets have order $o(H)$.

We now come to the major theorem:

Lagrange's Theorem:

Given G a finite group, and H a subgroup of G , $o(H)/o(G)$.

proof: Let there be k distinct right cosets of H in G . Since no two distinct right cosets have any common element, and every element is in some coset, then since each coset has exactly $o(H)$ elements, we have
 $ko(H) = o(G)$ and so, $o(H)/o(G)$.

Q.E.D.

Lagrange's Theorem is one of the fundamental theorems in group theory and is one of the most useful.

Def: If G is a finite group and $a \in G$, the order of a , is the least positive integer m , such that $a^m=e$. $o(a)=m$.

Lemma 7: If G is a finite group, then $o(a)/o(G)$.

proof: The set $e, a, a^2, a^3, \dots, a^{o(a)-1}$ is a group of order $o(a)$. The lemma then follows from Lagrange's Theorem.

Lemma 8: If G is a finite group and $a \in G$ then $a^{o(G)}=e$.

proof: From Lemma 7, $o(a)/o(G)$, so $o(a)k=o(G)$. So $a^{o(G)}=(a^{o(a)})^k=e^k=e$.

Lemma 9: Let G be the set of integers less than some specified integer n and relatively prime to n , under the operation of multiplication (mod n). Then G forms a group.

proof: G is just the reduced residue system mod n . Try and prove that it does form a group.

And now,

Euler's Theorem: If n is an integer, and a some integer relatively prime to n , then,

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

proof: Using the group specified in Lemma 9, noting that $o(G)=\phi(n)$ we have, for any $a \in G$,

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

If n is a prime $\phi(n)=n-1$ and we again have Fermat's Theorem. Euler's Theorem, however, is more general and so more powerful.

We have achieved our goal of proving Fermat's Little Theorem in two ways, but more importantly we have taken a glimpse into two vast areas of mathematics. There is much, much more in both number theory and group theory which are fascinating.

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Distance From a Point to a Line

In math class we learned a formula for finding the distance from a point to a line. Since the proof was rather involved, we will give a much shorter one now.

Let the line be

$$\frac{x}{a} + \frac{y}{b} = 1$$

and the point (x_1, y_1)

Hence the line cuts the x axis at a, and the y axis at b. From (x_1, y_1) make lines parallel to each of the axes and let them intersect the given line. It is easy to see that we have similar triangles. By area

$$\frac{1}{2} (ka)(kb) = \frac{1}{2} zk\sqrt{a^2+b^2}$$

or

$$z = \frac{kab}{\sqrt{a^2+b^2}}$$

Now we know that $ka = |x_1 - x_0|$

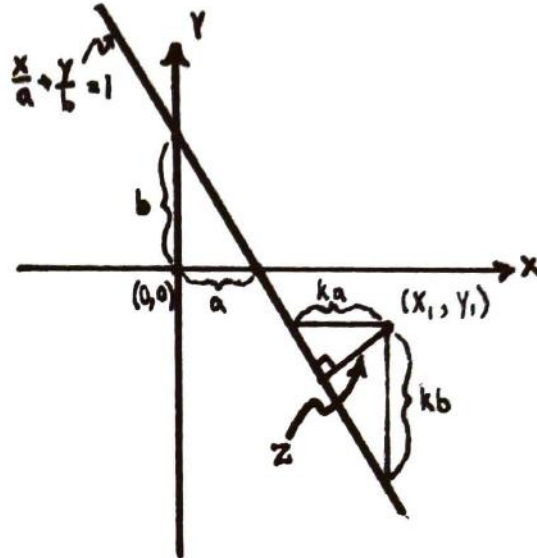
where x_0 is the x coordinate of

the point on the given line with y coordinate of y_1 . It follows that

$$kab = |bx_1 + ay_1 - ab|$$

or

$$z = \frac{|bx_1 + ay_1 - ab|}{\sqrt{a^2+b^2}}$$



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Problems

1. Prove: Any perfect square greater than 10 has at least two distinct digits.
2. Prove: The number of people in the world who have shaken hands an odd number of times is even.
3. Find an arithmetic progression, none of whose terms is a perfect k -th power, for any integer k .
4. $(\)x^4 + (\)x^3 + (\)x^2 + (\)x + (\) = 0$. Prove that if the parentheses are filled in, using each of the numbers: 17, 2, -20, 5, -4, each used once, the resulting polynomial has a rational root.
5. A strange amnesty is granted to the prisoners of cell block A. At first, all the cells are locked. Their condition can be changed from locked to open or back again by turning a key in the lock once. The jailor first opens every cell. (There are exactly n cells each numbered with the integers from one to n .) Then he turns the key in every cell whose number is a multiple of 2. Then, he turns the key in every cell whose number is a multiple of 3, ... and so on, ..., in every cell whose number is a multiple of n . Who is free to go?
6. Find the six non-negative integers that are equal to the sum of the cubes of the digits of their base ten representation.
7. Find a three digit decimal numeral "ABC" which is equal to $A! + B! + C!$.
8. If the sides of a right triangle are in arithmetic progression, prove that they must be in the ratio 3:4:5.
9. If $2\log(x-3y) = \log(4x) + \log(y) - \log 2$, (all logs real) find $x:y$. (Only one answer!!)
10. Solve for all real (x, y, z) :

$$z^x = y^{2x} \quad 2^z = 2(4^x) \quad x + y + z + 16$$
11. In the 1969 World Series, after three games were played, the New York Mets were leading the Baltimore Orioles two games to one in the best of seven contest. Assume that the odds of either team winning any of the individual remaining games is 50:50. Your local bookie offers you 2:1 odds that the Mets will take the Series. Should you take the odds. that is, is your bookie underestimating the probability that New York will win? If so, what are the odds?
12. Towns A, B, C, D are located on the vertices of a one mile square. With only $(1+\sqrt{3})$ miles of road, can you connect the towns in such a fashion that it is possible to travel from any one town to any other without passing through a third.
13. In the number $A = 0.1234567891011121314...$ all the integers are listed in succession. Prove that A is irrational.
14. Find all pairs of integers whose sum equals their product.
15. Solve in whole numbers:
 - (a) $1! + 2! + 3! + \dots + x! = y^2$
 - (b) $1! + 2! + 3! + \dots + x! = y^z \quad (z > 1)$
16. You are given a pan balance and twelve coins, one of which is counterfeit. It is not known whether it is heavier or lighter, than genuine coins. In three weighings determine the counterfeit.
17. In problem 16, take 14 coins of which 7 are identical counterfeits. Prove that at least eight weighings are required.
18. Prove that the expressions $2x+3y$ and $9x+5y$ are divisible by 17 for the same set of integral values of x and y .
19. Does there exist a polynomial $f(x)$ such that $f(n)$ is prime for every integer n ?

Life

by Cole Whiteman

Martin Gardner's "Mathematical Games" column in Scientific American has earned a reputation for bringing a wide variety of oddities and pastimes, old and new, to the attention of the mathematical community.

But the item that has had perhaps the widest popularity and impact is the now-famous "LIFE" game--devised by Sir John Horton Conway of Cambridge University, and first described in Gardner's column of the October 1970 issue.

"LIFE" has been used as an educational tool to demonstrate properties of biological growth to grade-schoolers; it has consumed thousands of hours of computer time in the hands of programming addicts who have become hooked on the game; and it has prompted papers and research on cellular automata theory.

*****PLAYING THE GAME*****

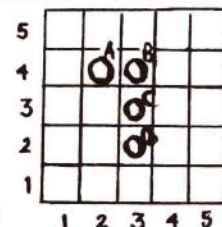
"LIFE" is played on a chessboard of arbitrary extent (infinite, if you like) which represents a cell culture medium. Each square on the board can either be "full" (occupied by a cell) or "empty". You start the game with a pattern of cells on the board. The pattern will change with time, in discrete steps called "generations", according to certain predefined rules. The fun is in watching the pattern of cells grow and develop as though it were a biological entity.

The rules governing the transition from one generation to the next are as follows. To determine what will happen to a given square, count up the number of occupied adjacent squares and plug that number, "N", into the following function:

```
IF N < 2 THEN any cell in the square under consideration will
                will die on the next generation--from "isolation",
                so to speak; ELSE,
IF N > 3 THEN any cell in the square will die from
                "overpopulation"; ELSE,
IF N = 3 AND the square is empty
                THEN there will be a "birth"--that square will be
                full on the next generation;
ELSE the state of the square remains unchanged for the next
      generation.
```

An example may suffice to make this process clear. Consider the initial pattern entitled "Generation 0".

First, let's check out the occupied squares. Cell A touches two cells (B and C): looking at our function, we see that nothing happens, so that location--(2,4) on the graph--will still be occupied on the next generation. Similarly, cell B touches 2 cells, so it will remain. Cell C touches



GENERATION 0

Now that you know how to play the game, what can you do with it?

(1) For one thing, you can try figuring out the life histories of various initial configurations in the hope of finding new organisms which do weird things. (I myself spent half of my freshman year--and a couple of reams of graph paper!--doing exactly that.)

For example, there's a design called a "Cheshire Cat" (see below) which shrinks into a "grin" after six generations and then stabilizes into a "paw print". There are "gliders"--such as the one shown below, which moves across the board over a period of time by reforming itself every 4 generations, displaced one square down and one square to the right. There exist configurations which travel and leave behind "debris", or which shoot out gliders, or which eat gliders, or eat other patterns, or explode . . . the possibilities are endless.

(2) Try variations on the rules--change the numbers in the transition function and see how the growth of various organisms is affected. Can you think of any reasons why Conway picked these numbers and not others?

If you're not satisfied, change the function altogether--maybe adding further possible states to the system.

(3) If you're a computer nut, try writing a program to carry out the LIFE process. It should be quite possible to compose a FORTRAN program for our 1130 which will read in an initial configuration from data cards and print out the life history of the organism for a given number of generations.

In fact, the operators' staff could really use a permanently stored interactive LIFE routine that could be "played" on the console directly.

*****"LIFE" IS JUST A CELLULAR AUTOMATON*****

(--as opposed to, let's say, a bowl of cherries)

On a theoretical level, Conway's model is an example of what is known among mathematicians as a cellular automaton system, which is built up from the following concepts, in the general case:

- (1) a cellular space--an array of locations, each of which is, at any time, in one of a finite number of states. For Conway's model, there are exactly two states--empty and full. A cellular space need not be two-dimensional, nor must it be rectangular; an array of hexagons, spheres or interlocking 37-dimensional bagels is just as acceptable.
- (2) a neighboring relation--which, for any location in the space, will yield a finite set of other locations, called the "neighbors" of the location under consideration. Defining the neighbors of a square as those squares adjacent to it, as in the Conway model,

is logical--but this too is not necessary. In checkers, for instance, a square has only four "diagonal" neighbors; and the neighbors of a knight in chess aren't even in contact with the knight's square. In all cases, however, the neighbors of a locations are those locations which will influence the one under consideration.

- (3) a transition function--which, given the states of a location, and its neighbors at time T, yields the state of the given location at time T+1.

One criterion for the set of possible cellular states is that the set must contain a "blank" or "quiescent" state, defined as follows--if a location Λ and its neighbors are "blank" at time T, then Λ is blank at time T+1. (Conway's "empty" state fulfills this condition. Try it if it's not immediately obvious.) This requirement precludes the possibility of systems in which cells can arise "spontaneously".

In light of these generalizations, Conway's model is comparatively simple.

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*****END



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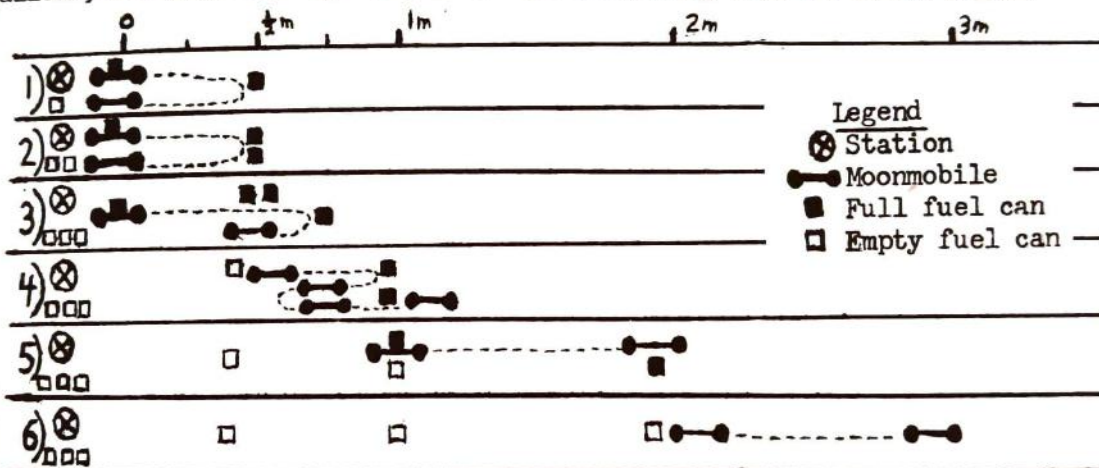
Moon Shot

by Francis Barany

You have just landed on the moon! Mission control instructs you to circumnavigate the moon using a moonmobile. Your moonmobile has a 10 gallon tank and is able to carry one and only one 10 gallon refueling can. This can may be dropped off onto the moon at strategic places, and may be picked up later. There is one gas station where you have landed. This station can refill the moonmobile tank until it holds exactly 10 gallons, no more and no less. The station also has an unlimited supply of 10 gallon refueling cans. The moonmobile has excellent mileage. It can travel $\frac{1}{5}$ of the moon's circumference using only 10 gallons.

Mission control has just given the go ahead signal, but directs you to use as little fuel as possible. What is the minimum amount of fuel needed to circumnavigate the moon?

In analyzing the problem, let us first take a simpler case, namely that the moon is a plane. If the moonmobile can travel m miles with ten gallons, then on the flat moon, it would take 10 gallons to travel m miles, and 20 gallons to travel $2m$ miles. However, it would require 60 gallons to travel $3m$ miles. Since the carrying capacity of the moonmobile is only 20 gallons, one must set up "stations" of refueling cans as shown below:



The following table shows the number of gallons of fuel required for m to $10m$ miles.

n (miles in m)	1	2	3	4	5	6	7	8	9	10
Y_n (gallons in 10 gal.)	1	2	6	22	86	342	1366	5462	21848	87390

Notice that $Y_{n+1} = 4(Y_n) - 2$ for all $n \geq 1$

Suppose we want a general rule for Y_n in terms of n .
We have

$$Y_2 = 4(Y_1) - 2$$

$$Y_3 = 4(4(Y_1) - 2) - 2$$

$$Y_4 = 4(4(4(Y_1) - 2) - 2) - 2$$

$$Y_n = \underbrace{4(\dots 4(Y_1) - 2)}_{n-1} \dots - 2$$

Simplifying we get

$$Y_n = 4^{n-1} (Y_1) - 2(4^{n-2} + 4^{n-3} + \dots + 1)$$

Substituting for Y_1 and summing the geometric series $4^{n-2} + 4^{n-3} + \dots + 1$, we obtain:

$$Y_n = 4^{n-1} - \frac{2(4^{n-1} - 1)}{3}$$

Combining terms under a common denominator we have

$$Y_n = \frac{3(4^{n-1}) - 2(4^{n-1}) + 2}{3} \quad \text{Simplifying we get} \quad Y_n = \frac{4^{n-1} + 2}{3}$$

Now we can tackle the original problem. On the moon, "stations" may be set up in two directions. Again, let the circumference of the moon be m , $2m$, $3m$, etc. miles. The problem may now be divided into two steps:

1) Placing a refueling can at the $m(\frac{n}{2} - 1)$ mark,

at the $m(\frac{n}{2} - 2)$ mark,

⋮

and at the $m(\frac{n}{2} - [\frac{n}{2}])$ mark.

2) Moving the moonmobile (in the other direction) a distance of $m(\frac{n}{2} + 1)$ miles. We already have a formula for this step.

The relationship between the number of gallons needed to travel from m miles to $10m$ miles is shown below.

n (miles in m)	1	2	3	4	5	6	7	8	9	10
Y_n (gallons in 10 gallons)	1	2	5	11	23	47	95	191	383	767

$$Y_{n+1} = 2(Y_n) + 1 \text{ for all } n \geq 2$$

By using the aforementioned method of solving for Y_n in terms of n , we get the general rule.

$$Y_n = 3(2^{n-2}) - 1$$

In the given problem n was equal to 5:

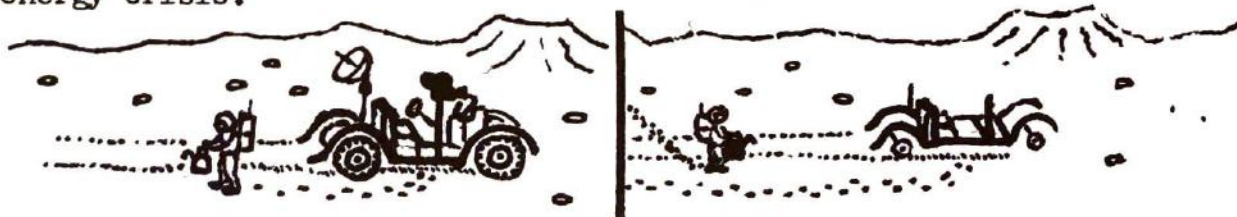
$$Y_n = 3(2^{(5-2)}) - 1$$

$$Y_n = 3(8) - 1$$

$$Y_n = 23$$

Hence 230 gallons are needed to circumnavigate the moon.

Hopefully the reader will apply the content of this article to allieviate the unpleasant effects of empty gas tanks caused by the energy crisis.



Simplified Slide Rule

by Danny Siegel

The slide rule is a valuable instrument in science and mathematics. Working on the principle of logarithms that is transformed into distances; i.e. $\log 2$ is .3010; on the C and D scales of a ten inch slide rule, 2 appears 3.01 inches from the left end of the scale.

If you don't already own a slide rule, borrow one. You are now ready to use this forerunner to the electronic calculator for fun and math.

First, we will learn to multiply. This is accomplished by using the C and D scales. To multiply, we set the one(1) on the C scale over one of the factors of the multiplication on the D scale. Next, we find the other factor on the C scale and look down to the D scale andVoila! We have our answer. Decimal places must be figured on paper or in one's head, because on the scales, 1.5 means 15 and 150 and 1.5×10^x (where x is a whole number greater than or less than 0).

EXAMPLE: $1.7 \times 2.8 = ?$

- A. Set the 1 on the C scale over 1.7 on the D scale.
- B. Find 28 (2.8) on the C scale. Read down to the D scale (You should get an answer close to 47.6. Watch your decimal points and remember that your answer is not perfectly accurate).

Now on to bigger and better things: division. Division is not worth much discussion, because it is the reverse of multiplication. You just find the dividend on the D scale, place the divisor on the C scale over it, and you find the quotient on the D scale under the 1 on the C scale.

For combined operations, such as $\frac{ab}{c}$, we first divide (a) by (c), then multiply by (b).

Example: $\frac{24.8 \times 1.54}{12.4} =$

- A. Set C and D scales to divide 24.8 by 12.4
- B. Read over to 1.54 on the C scale. Look below on the D scale and there is your answer (Remember those decimal places. Your answer should be near 3.08)

After working with your slide rule for a while, you may have noticed that your answers often appear off the D scale. This is easily remedied by using the 1 on the right hand side of the C scale.

We also use the slide rule to set proportions. All you have to do is set the original proportion and read off equivalent ones from anywhere on the scales.

Example: Set the proportion of one inch equals 2.54 centimeters.

- A. Set 1 on the C scale over 2.54 on the D scale.
- B. Anywhere you look on the C scale, you have the number of inches considered. Anywhere below on the D scale you have the metric equivalent.

Next we explore the CI scale. This scale contains the reciprocal of the C scale in decimal form. You read the CI scale from right to left. Example: Find the reciprocal of 2.5

- A. Look at 2.5 on the C scale. Look above this on the CI scale. You see a 4. This is actually a .4; $2.5 \times .4 = 1.00$

Squares and square roots can be worked out on the D and A scales. The A number is the square of the number on the D scale; and inversely the number on the D scale is the square root of the number on the A scale. The A scale is split into two logarithmic scales. To find squares look at the number on the D scale. Read up to the A scale and there is your square. To find the square root of a number with an odd number of digits, use the right part of the A scale: example $\sqrt{134}$

A. Find 134 on the left side of the A scale. Look down on the D scale. What do we find? Hint: It's between 11 and 12.

For cubes and cube roots, the K and D scales are used. The cube is on the K scale and the cube root is on the D scale. There are three scales on the K scale. The first is for numbers from 10^0 to 10^1 , the second: 10^1 to 10^2 , and so forth. This order repeats itself indefinitely. Example: $(22.3)^2$

A. Find 22.3 on the D scale. Read up on the K scale. There is your cube. Hint: It's between 10,000 and 20,000.

Although a slide rule does many things, the last one I will discuss is logarithms. This is quite simple. Just set the L and D scales with their ends meeting. Looking at the numeral on the D scale, we will find it's log on the L scale. The characteristic is determined by counting decimal places from the first non-zero digit of the numeral, either to the left (negative) or to the right (positive).

Example: Find the log of 18.

A. The log of 18 is found on the L scale to be .255

B. The characteristic is figured to be 1.

C. Therefore, $\log_{10} 18 = 1.255$

For further knowledge you can find information on the slide rule in almost any mathematics encyclopedia or reference book.

There are also numerous books and pamphlets on the slide rule itself.

INDUCTION ON THE MISSISSIPPI

In the short space of one hundred and seventy-six years the Lower Mississippi has shortened itself two hundred and forty-two miles. That is an average of a trifle over one mile and a third per year. Therefore, any calm person, who is not blind or idiotic, can see that in the Old Oolitic Silurian Period, just a million years ago next November, the Lower Mississippi River was upwards of one million three hundred thousand miles long, and stuck out over the Gulf of Mexico like a fishing rod. And by the same token any person can see that seven hundred and forty-two years from now the Lower Mississippi will be only a mile and three-quarters long, and Cairo and New Orleans will have joined their streets together, and will be plodding comfortably along under a single mayor and a mutual board of aldermen. There is something fascinating about science. One gets such wholesale returns of conjecture out of such a trifling investment of fact.

-- Mark Twain, Life on the Mississippi

CAREFULLY PLANNED PARENTHOOD

"The figure of 2.2 children per adult female was felt to be in some respects absurd and a Royal Commission suggested that the middle classes be paid to increase the average to a rounder and more convenient figure."

--Punch

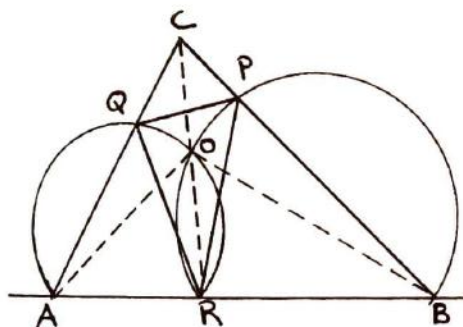
Schwarz's Proof

by Bruce Plamm

Hermann Amandus Schwarz (1843-1921) was a distinguished mathematician of the University of Berlin and one of the greatest contributors to modern function theory and analysis. He did not disdain to write on elementary subjects, and one of his papers treats the following problem: Given an acute angled triangle, to inscribe in it another triangle with the least possible perimeter. An inscribed triangle is a triangle with a vertex on each side of the original triangle. There is exactly one such triangle whose vertices are the foot-points of the altitudes of the given triangle. This triangle is called the altitude triangle.

Schwarz proved the minimum property of the altitude triangle by the method of reflection, with the help of the following theorem of elementary geometry: At each vertex, (see Fig. 1) P, Q, R , the two sides of the altitude triangle make equal angles with the side of the original triangle; this angle is equal to the angle at the opposite vertex of the original triangle. For example, the angles ARQ and BRP are both equal to angle C , etc.

To prove this preliminary theorem, note that $OPBR$ is a quadrilateral that can be inscribed in a circle, since $\angle OPB$ and $\angle ORB$ are right angles. Consequently, $\angle PBO$ equals $\angle PRO$, since they subtend the same arc PO in the circumscribed circle. Now $\angle PBO$ is complementary to $\angle C$, since CBQ is a right triangle, and $\angle PRO$ is complementary to $\angle PRB$. Therefore, the latter is equal to $\angle C$. In the same way, using the quadrilateral $QORA$, we see that $\angle QRA$ equals $\angle C$, etc.



This result leads to the following reflection property of the altitude triangle: Since, for example $\angle ARQ = \angle CQP$, the reflection of RQ in the side AC is the continuation of PQ , and vice versa; similarly for the other sides.

Now we can prove the minimum property of the altitude triangle. In the triangle ABC consider, together with the altitude triangle, any other inscribed triangle, UVW . Reflect the whole figure first in the side AC of ABC , then reflect the resulting triangle in its side AB , then in BC , then again in AC , and finally in AB . In this way we obtain altogether six congruent triangles, each with the altitude triangle and the other one inscribed. The side BC of the last triangle is parallel to the original side BC . For in the first reflection, BC is rotated clockwise through an angle $2C$, then through $2B$ clockwise; in the third reflection it is not affected, in the fourth it rotates through $2C$ counterclockwise, and in the fifth through $2B$ counterclockwise. Thus the total angle through which it has turned is zero.

Due to the reflection property of the altitude triangle, the straight line segment PP' is equal to twice the perimeter of the altitude triangle, for PP' is composed of six pieces that are, in turn, equal to the first, second, third side of the triangle, each occurring twice. Similarly the broken line from U to U' is twice the perimeter of the other inscribed triangle. This line is not shorter than the

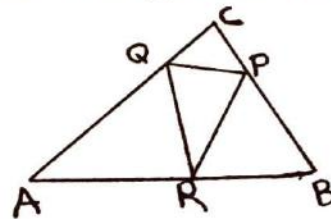
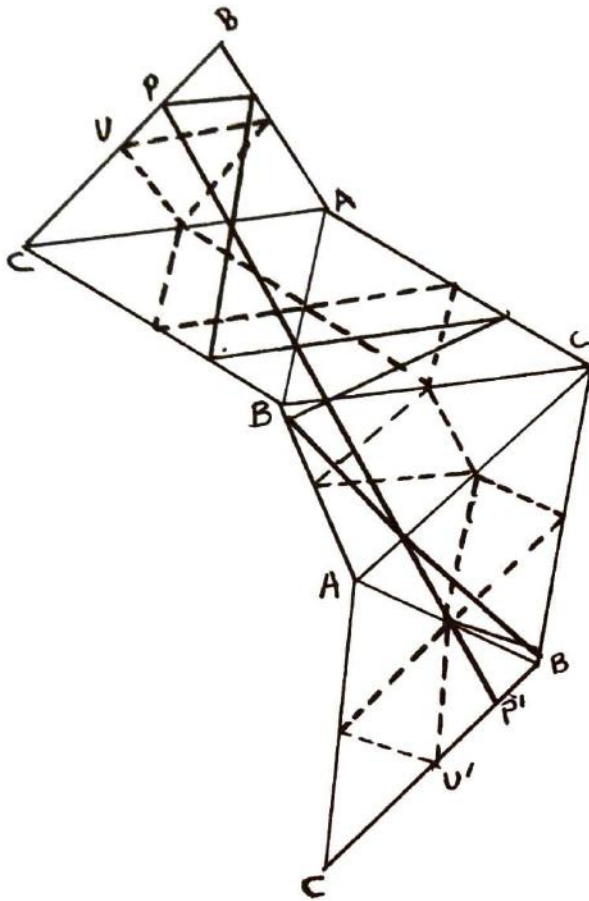
straight line segment $\overline{UU'}$. Since $\overline{UU'}$ is parallel to $\overline{PP'}$, the broken line from U to U' is not shorter than $\overline{PP'}$, and therefore the perimeter of the altitude triangle is the shortest possible for any inscribed triangle, as was to be proven! Thus we have, at the same time that a minimum exists and that it is given by the altitude triangle.

Perhaps the simplest solution of Schwarz's problem is the following:

The sum of the distances from two points P and Q to a line L is least for that point R on L which make the angles that PR and QR make with L equal. Assume that PQR , a triangle inscribed in triangle ABC solves the minimum problem. Then R must be the point on the side AB that makes $PR+QR$ a minimum, and therefore the angles ARQ and BRP must be equal; similarly for angles $AQR=CQP$ and $BPR=CPQ$. Thus the minimum triangle, if it exists must have the equal angle property used in Schwarz's proof, (see Fig. 3). It remains to be shown that the only triangle with equal angle property is the altitude triangle. (Moreover, we note that P, Q, R cannot coincide with any vertex of ABC).

We observe that if an inscribed triangle has the equal angle property, the angles P , Q , and R must be equal to angles A , B , and C , respectively. For

assume, say that angle $ARQ = \angle C + \epsilon$. Then since the sum of the angles of a triangle is 180° , the angle at Q must be $\angle B - \epsilon$ and at P , $A - \epsilon$, in order that the triangles ARQ and BRP may have the sum of their angles equal to 180° . But then the sum of the angles of the triangle CPQ is $(A - \epsilon) + (B - \epsilon) + C = 180^\circ - 2\epsilon$; on the other hand, this sum must be 180° so that $\epsilon = 0^\circ$. We have already noted that the altitude triangle has the equal angle property. Any other triangle with this property would have sides parallel to the altitude triangle; in other words it would be similar to the altitude triangle and oriented in the same way. As can be seen the Fig. , there is no other such triangle.



Σ

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Infinite Descent

by Paul Zeitz

Many of us have heard of mathematical induction. This exceedingly useful technique allows one to prove many statements involving sets of natural numbers, e.g., $1+2+3+\dots+n=1/2n(n+1)$, where n is a natural number, or $x^m x^n = x^{m+n}$, where m and n are natural numbers. (A natural number is a positive integer.)

The basic idea in mathematical induction is the easily accepted proposition that if S is a set of natural numbers, and

(i) $1 \in S$

(ii) $k \in S \implies k+1 \in S$ (k a natural number, and " $\implies$ " meaning implies) then S is the set of all natural numbers.

There is another technique, essential in the theory of diophantine equations (equations with only integer solutions, such as $x^2+y^2=z^2$), known as "Fermat's Method of Infinite Descent", discovered by the great 17th century French mathematician, Pierre de Fermat. The basic concept in infinite descent is as natural as the induction proposition, and can in fact be shown to be equivalent to the induction proposition. It says:

In every set of natural numbers, finite or infinite, there is a least element. (well-ordering property)

My illustrative example of using infinite descent in a proof will be a proof of one case of Fermat's Last Theorem. Fermat's Last Theorem is one of the most famous unsolved conjectures. It states that there does not exist any natural number solutions for the diophantine equation $x^n+y^n=z^n$, where $n \geq 3$. This theorem, while proven for many different n 's, has never been proven completely. For $n=2$, there are an infinite number of solutions, such as $(3,4,5)$, $(5,12,13)$, $(9,40,41)$, $(20,99,101)$, etc. The proof that I will discuss will be for the case $n=4$.

Before beginning the proof, we must have a few preliminary notions.

1. The greatest common factor of 2 integers, m, n is denoted by (m, n) . When $(m, n)=1$ the integers are said to be relatively prime.
2. If a solution of a diophantine equation contains only relatively prime integers, then the solution is called a primitive solution. E.g., $(3,4,5)$ is a primitive solution of $x^2+y^2=z^2$, while $(6,8,10)$ is not.

3. All primitive solutions of $x^2+y^2=z^2$ can be expressed in the form $x=u^2+v^2$, $y=2uv$, $z=u^2-v^2$, where $u>v>0$, $(u,v)=1$, and u and v are of opposite parity (one is even, the other odd, and vice versa). A proof of this can be found in any elementary text on number theory.

4. All perfect squares are of the form $4k$ or $4k+1$.

Now we can begin the proof. Since $x^4+y^4=z^4$ is equivalent to $x^4+y^4=(z^2)^2$, proving that $x^4+y^4=z^4$ has no natural number solutions. Also, proving the nonexistence of primitive solutions will of course imply the nonexistence of all solutions.

We shall employ indirect proof. We will assume that $x^4+y^4=z^4$ has at least one solution. Since these solutions are sets of natural numbers, there exists a solution (x,y,z) where z is the least possible of all the solutions. However, we shall show that there is no least z , hence no natural number solutions.

By 3., the primitive solutions of $x^4+y^4=z^4$ are $x^2=u^2-v^2$, $y^2=2uv$, $z^2=u^2+v^2$, $u>v>0$, $(u,v)=1$, u, v , opposite parity. If u were even, and v odd, then x would be of the form $4k-1$, which by 4. is impossible. Hence u is odd, v is even. We have $y^2=2uv$, and $1/4y^2 = (1/2y)^2 = u(v/2)$, $(u, v/2)=1$. We now have that the product of two relatively prime integers must be a square. This implies that both u and $v/2$ are themselves squares.

(Try a few examples if you aren't sure of the truth of this statement.)
 Thus we have, for some integers r, s : $u=r^2$, $v/2=s^2$, $y^2=4r^2s^2$? $y=2rs$,
 $(r,s)=1$, $r>0$, $s>0$, r odd. Now, r and s are relatively prime due to the
 fact that u and $v/2$ are relatively prime.

From before, we had $x^2=u^2-v^2$, or $x^2+v^2=u^2$. Substituting, we then
 get $x^2+4s^4=r^4$. Again by 3, we have integers p, q such that $x=p^2-q^2$,
 $2s^2=2pq$, $r^2=p^2+q^2$, $p>q>0$, $(p,q)=1$. Since $pq=s^2$, and $(p,q)=1$ then p and q
 are both squares, and so there exist both f and h such the $p=f^2$ and
 $q=h^2$. Since $r^2=p^2+q^2$ we get $r^2=f^4+h^4$.

But isn't this the same equation we started with? Not quite. This
 equation has different solutions than $x^4+y^4=z^2$, for $z=u^2+v^2=r^4+4s^4>r^4$
 or $z>r^4$, which gives us, $z>r>0$. Repeating this process, we get a new
 equation $r_1^2=f_1^2+h_1^2$ with $z>r>r_1>0$ and then $r_2^2=f_2^2+h_2^2$, with
 $z>r>r_1>r_2>0$.

We can thus obtain an infinite set of natural numbers $z>r>r_1>r_2>\dots>0$
 that "descends infinitely" and hence has no least element.

But we assumed that there would be a "least" positive solution.
 Hence there can be no solution. We have thus proven Fermat's Last
 Theorem for the case $n=4$.

For those interested in the foregoing, there is an extensive
 collection of literature on Fermat's Last Theorem, and other number theory
 topics. Vinogradov's Elements of Number Theory is an excellent book
 with few prerequisites. Students with mathematics above the M81 (first
 year college) level may find Niven's and Zuckerman's An Introduction to
the Theory of Numbers more suitable.

Solution to the Butterfly Problem

Let $QM=x$ and $MP=y$; AM and MB are each equal to a . We will
 agree to call the area of triangle RST , $k(RST)$. It is easy to see
 that

$$\frac{k(QCM)}{k(PEM)} \cdot \frac{k(QFM)}{k(PDM)} = 1$$

So

$$\frac{CM}{EM} \cdot \frac{CQ}{EP} \cdot \frac{\sin \alpha}{\sin \alpha} \cdot \frac{EM}{FM} \cdot \frac{MP}{MQ} \cdot \frac{\sin \gamma}{\sin \gamma} \cdot \frac{FM}{MD} \cdot \frac{FQ}{DP} \cdot \frac{\sin \beta}{\sin \beta} \cdot \frac{MD}{CM} \cdot \frac{MP}{MQ} \cdot \frac{\sin \delta}{\sin \delta} = 1$$

Simplifying

$$CQ \cdot FQ \cdot (MP)^2 = EP \cdot DP \cdot (MQ)^2$$

but

$$CQ \cdot FQ = AQ \cdot QB \text{ and } EP \cdot DP = BP \cdot AP$$

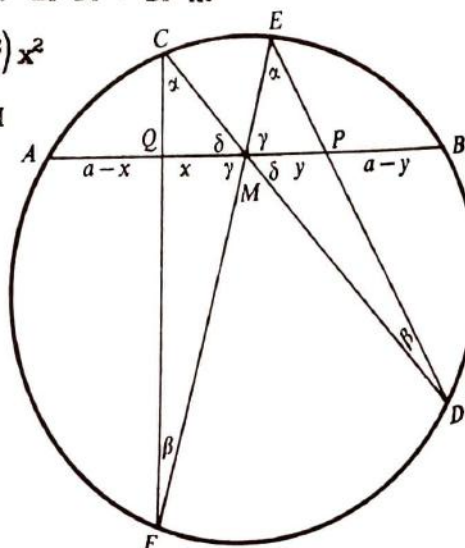
so

$$(a^2 - x^2)y^2 = (a^2 - y^2)x^2$$

or

$$x = y \text{ when } PM = QM$$

Solution due to
 STEVEN CONRAD of
 Benjamin N. Cardozo H. S.



Gödel's Incompleteness Theorem

by Eric S. Lander

1. Claim and Model. A logical question to run through the mind of a mathematician is "Is everything provable?", "Is it possible to neatly divide up the world into two parts - true things and false things-such that you can prove all true things and cannot prove any false things? So, if 'A' is true, 'not -A' is false and vice versa?"

Intuitively one wants to answer "yes", but as the startling Gödel's Incompleteness Theorem (1932) says, "It ain't so!" In this article, we shall follow the steps of the proof of this amazing claim. All rigor has been sacrificed in favor of expressing the main ideas. Interested readers can find much more complete proofs (or could fill in the gaps in this one) but it is the concept that is important. This is a theorem few people ever get to see.

The proof is by trade off and contradiction. We assume the opposite -- that we can prove everything. This is our goal. We shall reach a contradiction from it.

Along the way we shall trade off 'fuzzy' statements (like the theorem is now) for sharper, clearer equivalent statements until the contradiction becomes easy.

Since "everything" is a 'real world' term, it is hard to talk about it mathematically. We must trade off that notion for a clearcut mathematical notion -- a model. The physics student who neglects the minor details of air resistance, etc, and considers a falling block as a point is actually using a model, an approximation of reality. The quality of the approximation determines how applicable the answer obtained is. We shall try to make a 'good' model for "everything".

2. Everything is a set. One of the developments of mathematics in the last one hundred years is the notion of a set. I propose that everything may be thought of in terms of sets, that is, sets are a model of the "real world". Of course, I cannot prove this statement, for proofs are 'things' that take place within models, not between models and the worlds they are made to describe. At best, I can convince you of the plausibility of my claim. If not, if you deny that everything is a set, then I will not succeed in proving my claim of the previous section because you deny the applicability of the model. However, it is hard to think of something that is not a set.

Set theory has its own language, made up of symbols:

=	equality	~	not
∈	is a member of	⇒	implies
(	open parenthesis	⇔	is equivalent to
)	close parenthesis	∃	there exists
and	and	∀	for all
or	or	$x, x_1, x_2, \dots$	variables

Natural Numbers may be thought of as sets:

$$\begin{aligned}
 0 &= \emptyset \\
 1 &= \{\emptyset\} \\
 2 &= \{\emptyset, \{\emptyset\}\} \text{ or } 1 \cup \{1\} \\
 3 &= 2 \cup \{2\} \\
 &\vdots \\
 n+1 &= n \cup \{n\}
 \end{aligned}$$

Ordered Pairs:

$(a,b) = \{a, \{a,b\}\}$ (why not $\{a,b\}$?)

And, generalizing, triples, quadruples, ..., K-tuples, and sequences may be thought of as sets.

Addition: may be thought of as a set of ordered triples "+"

1. $(0,0,0) \in +$
2. $(1,0,1) \in +$
3. $(0,1,1) \in +$
4. If $(m,n,p) \in +$ then $(n,m,p) \in +$
5. If $(m,n,p) \in +$ then $(m+1,n,p+1) \in +$
6. If $(m,n,p) \in +$ then $(m,n+1,p+1) \in +$
7. Nothing else except as forced by 1-6.

We can assign numbers to each of the symbols of Set theory:

1 = -, 2 = $\in$, 3 = (, 4 =), ...

So that our code for $\exists x(x=x)$ would be:

10-13-3-13-1-13-4

Rather than a string of numbers we could assign the sentence the number:

$2^{10}3^{13}5^{37}7^{13}11^{13}13^{17}4$

(where the primes are consecutive). Since each positive integer is factorable uniquely into primes, there is no ambiguity in the representation. Hence, to each sentence we can assign a Godel Number.

Not everything in set theory is a sentence. For example, $\forall()xx$ is meaningless. However, one can, with a little time and thought devise some simple rules to identify what combinations of symbols make sentences. (Rules so simple, a machine could follow them.)

3. Axioms. The study of geometry, at one time, was thought of as a description of the world, as ultimate truth. The "truth" of Euclid's Fifth Postulate ("through a point there exists exactly one parallel to some given line") was debated. Nowadays, we have changed our notion of geometry (though most high schools texts refuse to acknowledge it) to that of a "game" - - - - We arbitrarily decide on some statements that we call "true". (axioms). We then proceed from these initial statements, and some rules of logic, to prove some more "true" statements (theorems). Euclidean Geometry (alias High School Geometry, alias Plane Geometry, alias Geometry) is just one of the many possible games -- though it seems to be the most popular and by far, the most developed. Hence, the Parallel Postulate can be accepted or denied and neither system is "right".

Similarly, in set theory we may choose our axioms however we desire. However, without some agreed-on axioms your system might look nothing like what is commonly called "set theory". In section 2 we said the natural numbers could be defined in terms of ϕ -- what if your system does not have an empty set?! Hence we shall agree on four axioms. After that, you may add as many more as you wish!

S1-There exists an empty set or

$\exists x \forall y (\sim y \in x)$

S2-Two sets with the same elements are equal.

(Try to write in set theory language).

S3-The union of two sets exists.

S4-For any x there exists $\{x\}$.

4. Register Machine. With the sophistication of the last section, we can restate our assumption, the statement we shall contradict as:

"There exists a set of axioms including S1-S4 such that for every sentence A we can derive either A or $\sim A$, but not both."

Claim: If this is so, then there exists a mechanical method for telling if a statement of set theory is "true"!!

proof: In section 2 we assigned Godel numbers to each sentence of set theory. Since a proof is merely a finite collection of sentences, we can assign each "proof" a Godel number, as follows:

Suppose we have a five line proof whose sentences have the Godel numbers g_1, g_2, g_3, g_4 , and g_5 , respectively. Then the proof has Godel number:

$$\begin{matrix} g_1 & g_2 & g_3 & g_4 & g_5 \\ 2^1 & 3^2 & 5^3 & 7^4 & 11^5 \end{matrix}$$

Proofs, of course, must "make sense". This abstract concept may be stated in a very mechanical way.

- (1) every statement is an axiom OR
- (2) every statement is a consequence, by modus ponens, of two preceding statements. Modus Ponens is the rule of logic that says if $(A \Rightarrow B)$, and A , then we can derive B . (If I am good I will get ice cream, and I am good, derives — I will get ice cream.) (It does not matter that we only allow the use of modus ponens, because Godel's Completeness theorem proves that modus ponens is the "only" rule of logic. We shall not prove this here.)

Now, the mechanical method for determining if a sentence A is true is simply:

- (1) Write a computer program to check each "proof", in order of increasing Godel numbers, to see if it "makes sense".
- (2) If a proof "makes sense", see if its last statement is A or $\sim A$. If not, continue. If so, stop, for we have a proof which tells us whether it is A or $\sim A$ which is true.

That eventually, at some finite time, A or $\sim A$ comes out of our computer is assured by our assumption that everything or its negation was provable. We will reach our contradiction by showing that the mechanical method cannot exist, however!!

It should be intuitively obvious to someone who is familiar with computer programming that it is possible to program a machine to do what we required above. Even a person who has no working knowledge of computers should recognize that the steps are essentially mechanical.

In fact, you do not need a high powered computer language to do the job. A very simple machine will suffice. Consider a machine with five registers labelled A, B, C, D, E , each of which may be used to store arbitrarily large(+) integers in them. Our language consists of five statement types:

$V \leftarrow V+1$ "add 1 to the contents of some register V "
 $V \leftarrow V-1$ "subtract 1 from the contents of register V if something is there"
 $\rightarrow (V=0)/k$ "If there is nothing in register V go to line k of the program. Otherwise continue."
 $V \leftarrow \square$ "Put input that is fed in, in register V "
 $\square \leftarrow V$ "Print the contents of register V "

(The $\square$ may be thought of as a door, letting numbers in and out.)
 (This language is a subset of APL.)

It turns out that this machine, which we shall call a Register Machine (henceforth R.M.), can do everything an IBM/360 can do, although the programs needed will be considerably longer. Addition, subtraction, input, output, and comparison are actually the only thing machines do, anyway. Our technique of Godel numbers makes the limitation to five register no problem, since we can "encode information" in an integers. (Try to convince yourself of the power of the R.M. by writing programs to multiply, or test if a number if prime — everyone in a computer class should at least try.)

6. Programs as Sentences in Set Theory. Any programmer who has tried to de-bug a program knows how to "Play Computer". That is, follow step by step instructions of your program to see why it has an infinite loop or does something it shouldn't. Let's examine a R:M: program:

[1] $A \leftarrow 0$	
[2] $\rightarrow (A=0)/7$	[6] $\rightarrow (B=0)/2$
[3] $A \leftarrow A-1$	[7] $0 \leftarrow B$
[4] $\rightarrow (A=0)/4$	
[5] $A \leftarrow A-1$	

Try this for $A=0,1,2,3$, and 4 and convince yourself that if A is even the program prints "0" and if A is odd it never halts (infinite loop). For $A=2$ let's follow the action:

Step	Contents of A	of B	Step	Contents of A	of B
1	2	0	5	11	0
2	2	0	6	0	0
3	1	0	7	0	0
4	1	0		0	0

Saying that "the program halts for input of some A " is equivalent to saying that "there exist a finite sequence of triples (like the one above) such that

1. The first triple is $(1,A,0)$
2. The last is $(7,0,0)$
3. $(1,m,n)$ is followed by $(2,m,n)$
- ...etc...

following the instructions of the program. Now, triples and sequences are part of set theory. Hence asking the question "does some particular program P halt for input A " is the same as asking "is a certain sentence of set theory (one beginning: there exists a finite sequence of) true?" In the last section, we saw that there is a mechanical way (a program, that is) for determining if a sentence of set theory is true. Hence

There must be a program that determines if any program P will halt for input A (we consider only 1-input programs)

As our last step in our proof by contradiction we shall show this is impossible!

7. The Halting Problem. Using the same method as for set theory we can assign Godel numbers to each and every R:M: program. Let us list ALL programs in order of increasing Godel number.

$P_1, P_2, P_3, \dots, P_n, \dots$

Suppose that there does exist some program which can decide if any P_k halts for input A . Let $P_k(A)$ be the output of program P_k with input A , if it halts. Then surely there exists a program to evaluate

$$Q(n) = \begin{cases} P_n(n)+1 & \text{if } P_n \text{ halts for input } n \\ 1 & \text{if } P_n \text{ does not halt for input } n \end{cases}$$

- by (1) determining if P_n halts for input n
 (2) "running it" and adding "1" to its output if it does, or, printing "1" if it does not

However, (here's the trick!!) Q is itself a R.M. program so it is on our list of all programs above. That is, Q is P_k for some k .

QUESTION: What is $Q(k)$?

- (1) since $Q=P_k$ we have $Q(k)=P_k(k)$
- (2) But, by definition of Q above, we have
 $Q(k)=P_k(k)+1$!!! (since P_k halts)

CONTRADICTION !!!!!!!

Hence:

- (1) There is no halting program.
- (2) You cannot mechanically decide the truths of set theory
- (3) It is not the case that there exists a set of axioms including S_1 - S_4 , such that using modus ponens alone you can derive, for every sentence A either A or $\neg A$, but not both.
- (4) You can't prove everything !

For a more rigorous treatment the reader is commended to Godel's Proof by Nagel and Newman.

Research Problems

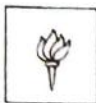
In this new section are presented some difficult and challenging problems that should serve as food for thought for the teacher and student alike. Some are unsolved (partial results due to Stuyvesantians are listed). It is hoped this section will inspire research and papers and will be continued in the future.

1. A sequence is an ordered collection of numbers $a_1, a_2, a_3, \dots, a_n$. A subsequence is obtained by throwing out terms of a given sequence, keeping in order those terms which remain. [For example, $(1, 2, 5)$ is a subsequence of $(1, 2, 3, 4, 5, 6)$] We call a sequence monotone increasing (decreasing) if each term is greater (less) than or equal to its predecessor.

Find a function $f(n)$ such that every sequence of $f(n)$ terms has a monotone subsequence of m terms (increasing or decreasing).
($f(1)=1, f(2)=2, f(3)=5, f(4)=12, \dots$)

2. Can the product of k consecutive integers ever be a perfect square ($k>1$)? E.S. Lander has answered the question negatively for $k=2, 3, 4, 5$ but a complete general solution is lacking. The case $k=2^n$ looks hopeful.
3. What integers can be expressed as the sum of consecutive positive integers? ($3=1+2, 27=7+8+9, 16$ cannot be expressed in this way) In how many ways can the integer N be so expressed?
4. Given a long strip of paper with the number "1" written on each end. Between the 1's write a "2". Then between the 1 and 2 write a "3". Continue in this fashion by on each pass, placing the sum of two integers between them. Each integer surely appears only finitely often, say, $g(n)$ times. Find $g(n)$.
5. Consider a triangular array of checkers as on the right. Remove one checker and discard it. Then proceed to jump one checker over another, removing the jumped checker, (remember only jumps) until only one remains. Is this always possible? How? Generalize to other shapes..

F. Barany has worked out strategies for some cases of the equilateral triangle.



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March 18, 1974.

To the Students of Stuyvesant High School:

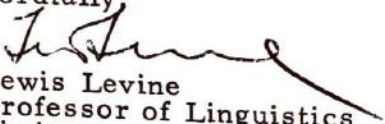
After a lapse of many years, I recently got a chance to go through several recent issues of Math Survey. Reading the journal affirmed my pleasure in being an alumnus.

It also gave me an urge to write this, to suggest that you consider doing your undergraduate work at New York University. Many of you are committed to specializing in mathematics or the physical sciences in college and do not need me to remind you of the reputation of our mathematics department, and Courant Institute of Mathematical Science, or of the reputations of our Physics, chemistry and biology departments. However, many of you want to use college to go in other directions.

I'd like to address myself for a moment to the members of this group. I'd like to call to your attention the fact that although our College is small, the range of subjects one can explore here is most broad. For example, we are among the few institutions in the country to offer undergraduates a full major in linguistics, a subject within which one can study language structure and use from points of view which extend from the mathematical to the sociological to the psychological. We offer generally excellent programs in the classical languages and history; in modern languages and literatures; in English; in history; in music; in anthropology, sociology and the other social sciences; and, we offer strong pre-medical and pre-law programs.

Regardless of the major you propose to take, some of you will be interested in our Early Admissions policy. Under that program you will need to meet only our regular admissions requirements, will be eligible for scholarship and will be a regular member of the freshman class of the year you enter. If you want specific information about our programs and admission applications or, if you simply want a chance to talk to professors of fields you want to explore, before thinking concretely about college, write to Dean Philip Mayerson, 910 Main Building, New York University College of Arts and Science, Washington Square New York, New York 10003. Or phone for an appointment (598-2425).

Cordially,


Lewis Levine
Professor of Linguistics
Chairman

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for the Advancement
of Science and Art

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Administrative
Offices

May 1, 1974

To the Underclassmen of Stuyvesant High:

This is a neighborly salute to Stuyvesant High School and its interesting publication, Math Survey, and a reminder to its readers that Cooper Union has degree offerings of special interest to mathematics students.

Our freshman class entering in September, 1974, has already been selected and some of Stuyvesant's current seniors are enrolled. But for today's Stuyvesant juniors, sophomores, and freshmen there is still plenty of time to stroll over to nearby Cooper Union and explore the courses of study we offer beginning in September, 1975.

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Write the Admissions Office for a catalog and application form--or, even better, walk over to Cooper Union and have a talk with me or a member of my staff. We'll be glad to see you.

Admissions Office
Room 202, Foundation Building
The Cooper Union
Cooper Square
New York, N.Y. 10003

Sincerely,
Herbert Liebeskind
Herbert Liebeskind
Director of Admissions

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